MUHAMMAD RAFI ANDRIANTO

JUNIOR DATA SCIENTIST | PREDICTIVE MODELING, COMPUTER VISION & BUSINESS ANALYTICS

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SUMMARY

Data Science & Machine Learning graduate with hands-on experience in Python, predictive modeling, exploratory data analysis, and business-focused analytics. Built and evaluated machine learning solutions for e-commerce delivery risk and hotel booking cancellation, including threshold optimization, SHAP-based interpretation, and financial impact analysis. Seeking a Junior Data Scientist role where predictive analytics and data-driven business insights can support operational and commercial decisions.

SKILLS

Technical Skills: Python, SQL (MySQL), Pandas, NumPy, Scikit-Learn, TensorFlow, PyTorch, MediaPipe, Exploratory Data Analysis, Data Cleaning, Data Visualization, Predictive Modeling, Time-Series Forecasting, Computer Vision, Classification, Model Evaluation
Tools: Jupyter Notebook, Tableau, Looker Studio, Matplotlib, Seaborn, OpenCV, Git, n8n
Soft Skills: Problem Solving, Adaptability, Analytical Thinking, Proactive Communication
Languages: Indonesian (Native), English (Fluent), Japanese (Intermediate)

EDUCATION

Ritsumeikan University, Osaka, Japan 2021 – 2025
Bachelor of Engineering, College of Information Science and Engineering

- GPA: 4.4/5.0
- Relevant Coursework: Artificial Intelligence, Software Engineering, Data Structures and Algorithms, Database, Computer Networks, Web Information Engineering

Purwadhika Digital Technology School, BSD, Indonesia 2026
Data Science and Machine Learning

- Core training: Python, SQL, Data Analysis, Machine Learning, Data Visualization, Business Analytics

PROJECTS

E-Commerce Late Delivery Prediction

Python, Scikit-learn, SQL, Pandas, Tableau

- Addressed operational bottlenecks across 100,000 orders where late deliveries severely degraded customer satisfaction, causing average review scores to plummet from 4.21 to 2.26 stars and directly threatening the platform's growth and reputation.
- Merged 9 complicated relational e-commerce datasets to perform statistical profiling, subsequently building a tuned LightGBM classification pipeline with SMOTE resampling and SHAP interpretability to proactively screen high-risk late delivery orders.
- Conducted interpretability analysis that provides the internal operations team with proactive risk-mitigation strategies to protect platform reputation, generating R$15,793.24 in net value and increasing order review scores from 3.01 to 3.98 stars.

Hotel Booking Cancellation & Revenue Loss Prediction

Python, Scikit-learn, SHAP, Pandas

- Addressed substantial financial leakage and inventory misallocation caused by high hotel booking cancellation rates, which led to tens of billions of IDR in unrecouped operational losses across reservation pipelines.

- Developed a hyperparameter-tuned XGBoost predictive classification model benchmarked across 11 algorithms, conducting threshold optimization and cost-benefit analysis to target prospective booking cancellations.
- Conducted interpretability analysis and threshold optimization to translate key cancellation drivers into actionable overbooking and policy recommendations for executive stakeholders, preserving IDR 5,411,962,956 in room revenue and generating a 1,512.57% ROI on the modeled outreach campaign.

Microplastic Accumulation Prediction

Python, TensorFlow, Keras, Scikit-learn

- Addressed the challenge of monitoring complex microplastic pollution trends across diverse locations, replacing manual sampling with a predictive machine learning framework to identify regional accumulation drivers.
- Engineered and evaluated Random Forest regression and Long Short-Term Memory (LSTM) deep learning time-series models on log-transformed temporal datasets, tracking training vs. validation loss curves to ensure model convergence and prevent overfitting.
- Achieved strong predictive performance with the Random Forest model by accounting for the vast majority of accumulation variance, enabling environmental teams to accurately forecast spatial concentration levels and target priority cleanup hotspots.

Skateboarder Performance Analysis System

Python, TensorFlow, Mediapipe

- Addressed the absence of accessible, objective biomechanical feedback for skateboarders struggling to master complex athletic tricks and identify subtle execution errors.
- Architected a hybrid computer vision pipeline using dual 3D CNN architectures to detect trick initiation phases, alongside MediaPipe pose estimation to extract and compare joint keypoint trajectories against ideal benchmark video executions.
- The system provided an automated visual analysis tool that highlights body joint trajectory discrepancies, helping skateboarders independently self-correct technique, shorten learning curves, and prevent impact-related physical injuries.

ACHIEVEMENTS

Ritsumeikan University Tuition Reduction Award 2023 & 2024

- Acknowledged for being among the top 20% of students and granted a 50% tuition reduction for the academic year 2023 & 2024.

Ritsumeikan University International Students Assistance Scholarship 2022

- Received a grant of ¥250,000 as a testament to academic excellence and contributions as an international student.

ORGANIZATIONS

Ritsumeikan University Google Developers Club 2022 – 2023

- Supported AI workshops designed to introduce new students to machine learning concepts.
- Helped deliver introductory learning content for students beginning their exploration of artificial intelligence and machine learning.

CERTIFICATIONS

Purwadhika Digital Technology School 2026
Data Science & Machine Learning